



ElectronicsTutorials

CAPACITORS & CAPACITANCE

For Students, Professionals
and Beyond

eBook 5

ASPENCORE

WWW.ELECTRONICS-TUTORIALS.WS

TABLE OF
CONTENTS

1. Introduction	1
2. The Capacitance of a Capacitor.	2
3. Multiplate Capacitor	3
4. Units of Capacitance	3
5. Standard Capacitance Values	4
6. Energy Stored in a Capacitor	4
7. Capacitor Types	4
8. Capacitor Characteristics	6
9. Capacitor Colour Coding	7
10. Capacitors in Parallel	7
10. Capacitors in Series	8
11. RC Charging Circuit	9
12. RC Discharging Circuit.	10
13. Capacitors in AC Circuits	11

Our Terms of Use

This **Basic Electronics Tutorials eBook** is focused on capacitors with the information presented within this ebook provided “as-is” for general information purposes only.

All the information and material published and presented herein including the text, graphics and images is the copyright or similar such rights of Aspencore. This represents in part or in whole the supporting website: **www.electronics-tutorials.ws**, unless otherwise expressly stated.

This free e-book is presented as general information and study reference guide for the education of its readers who wish to learn Electronics. While every effort and reasonable care has been taken with respect to the accuracy of the information given herein, the author makes no representations or warranties of any kind, expressed or implied, about the completeness, accuracy, omission of errors, reliability, or suitability with respect to the information or related graphics contained within this e-book for any purpose.

As such it is provided for personal use only and is not intended to address your particular problem or requirement. Any reliance you place on such information is therefore strictly at your own risk. We can not and do not offer any specific technical advice, troubleshooting assistance or solutions to your individual needs.

We hope you find this guide useful and enlightening. For more information about any of the topics covered herein please visit our online website at:

www.electronics-tutorials.ws

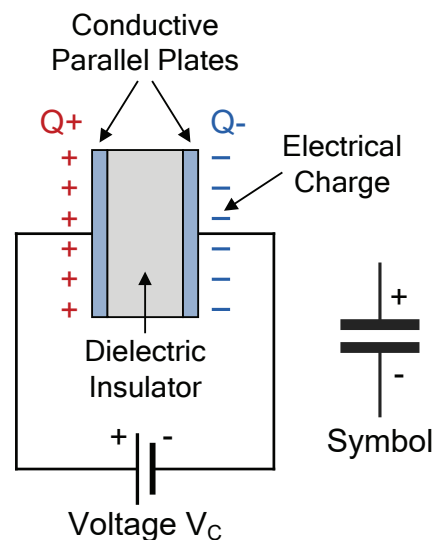
1. INTRODUCTION

Capacitors, (C) are simple passive devices which have the ability or “capacity” to store energy in the form of an electrical charge. Capacitors consists of two or more parallel conductive metal or foil plates which are not connected or touching each other. These conductive plates are physically separated from each other by a distance, (d) using either air or some form of insulating material called the **Dielectric**.

When a dielectric insulating material such as paper, mica, ceramic or plastic is inserted between the plates of a capacitor, it allows the capacitor to store more charge for a given potential difference across it.

Thus a capacitors ability to become charged by a voltage and then hold that charge indefinitely allows capacitors to be used in many electrical and electronic circuits in a variety of ways, from smoothing out fluctuations in voltage power supply levels to timing and filter circuits when used in conjunction with a resistor.

FIGURE 1. THE STRUCTURE OF A CAPACITOR



Capacitors vary from each other in shape and size, but the basic configuration for a parallel plate capacitor is of two conducting plates carrying equal but opposite charges.

The symbol used in schematic and electrical drawings to show a capacitor is generally two parallel lines as shown in Figure 1.

When the plates of an uncharged capacitor are firstly connected to a DC supply voltage, it takes some time for the charge (in the form of electrons) on the plates to reach their full intensity.

During the charging process, electrons move from one plate to the other one, producing a positive charge **+Q** on one plate, and a negative charge **-Q** on the other plate.

When a sufficient amount of electrical charge, **Q** (measured in units of coulombs) has been transferred from the source voltage to the capacitors plates, the voltage across the plates, **V_C** will be equal to the source voltage, **V_s** and the flow of electrons will cease.

The material used to separate the two parallel plates of a capacitor from each other is called the Dielectric

Thus a potential difference is created across the plates, with the positively charged plate at a higher potential than the negatively charged plate but as the two plates have opposite but equal charges, the difference in charge across the capacitor will be zero. That is: **+Q = -Q**.

The charge developed across the capacitors plates is not instantaneous but builds up slowly at a rate that depends on the capacitance value of the plates, the greater the capacitance, the slower the rate of change of voltage on the plates.

The capacitance, (**C**) value of a capacitor is an expression of the ratio between the amount of charge flowing and the rate of voltage (**ΔV**) change across the capacitors plates. A capacitance of one farad, (**F**) represents a charging current of one ampere when there is a voltage, **V** increase or decrease at a rate of one volt per second.

Then one coulomb of charge exists when a capacitance of one farad is subjected to one volt of potential difference and for a parallel plate capacitor the ratio of **Q ÷ V** is a constant called the capacitance, **C** as shown.

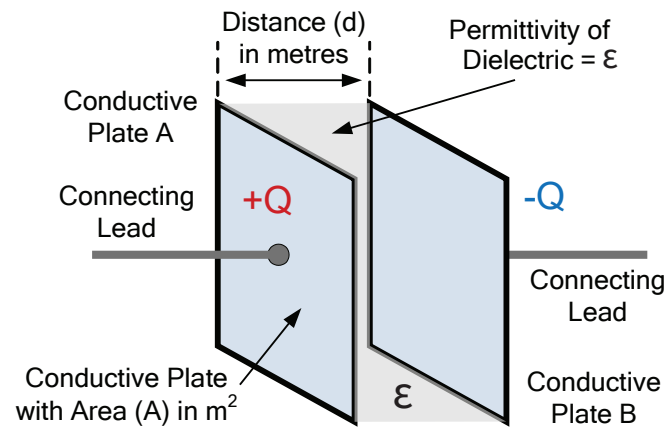
$$Q = V \times C, \quad C = Q \div V, \quad V = Q \div C$$

Where: **V** is in Volts, **Q** is in Coulombs and **C** is in Farads. Note that when **C** is given in microfarads, (**μF**) and **V** is given in volts, the charge **Q** will be in micro-coulombs (**μC**). Also increasing **C** while keeping the charge **Q** the same, will result in a decrease of the potential difference, **V** across the plates.

2. THE CAPACITANCE OF A CAPACITOR

The capacitance of a parallel plate capacitor is commonly defined as being proportional to its area, A in metres² of the smallest of the two plates. It is also inversely proportional to the distance or separation, d (i.e. the dielectric thickness) given in metres between these two conductive plates. Consider the example shown in Figure 2.

FIGURE 2. A PARALLEL PLATE CAPACITOR



The generalised equation for the capacitance value of a standard parallel plate capacitor in air is given as:

$$C = \epsilon(A/d)$$

Where:

“ ϵ ” is the “absolute permittivity” of the dielectric material being used to separate the two conductive plates.

“ A ” is the area squared of the smallest plate.

“ d ” is the linear distance separating the two conductive plates.

Thus we can see that the capacitance (C) of a capacitor is determined by its physical parameters. But as well as its physical dimensions, another factor which affects the overall capacitance of a capacitor is the “Permittivity” (ϵ) of the actual dielectric material being used.

The permittivity of a vacuum, ϵ_0 also known as the “permittivity of free space” has the value representing the constant 8.84×10^{-12} Farads per metre. But the plates of a capacitors are separated by a material of some kind such as air or a solid insulator rather than the vacuum of free space.

The dielectric constant, K is the ratio of the permittivity of the material used to that of free space - a vacuum

When calculating the capacitance of a capacitor, we can consider the permittivity of air, and especially of dry air, as being the same value as a vacuum as they are very close and this is the base value to which all other dielectric materials are referenced too.

The factor by which the dielectric material, or insulator, increases the capacitance of the capacitor compared to air is known as the **Dielectric Constant, k** and a dielectric material with a high dielectric constant is a better insulator than a dielectric material with a lower dielectric constant. Thus the dielectric constant of a vacuum is one, ($K = 1$). Dielectric constant is a dimensionless quantity since it is relative to free space.

If we take the permittivity of free space, ϵ_0 as our base level and make it equal to one, when the vacuum of free space is replaced by some other type of insulating material, their permittivity of its dielectric is referenced to the base dielectric of free space giving a multiplication factor known as “relative permittivity”, ϵ_r .

The actual permittivity or “complex permittivity” of a dielectric material between the two plates is therefore the product of the permittivity of free space (ϵ_0) and the relative permittivity (ϵ_r) of the material being used. This is given as: ($\epsilon = \epsilon_0 \times \epsilon_r$). This gives us a final equation for the capacitance of a capacitor as being:

$$C = \frac{\epsilon_r \epsilon_0 A}{d} = \frac{k \epsilon_0 A}{d} \text{ Farads}$$

Typical units of dielectric constant, **k** for common materials are: A Pure Vacuum = 1.0, Free Air = 1.0006, Paper = 2.5 to 3.5, Glass = 3 to 10, Mica = 5 to 7, Wood = 3 to 8 and Metal Oxide Powders = 6 to 20, etc.

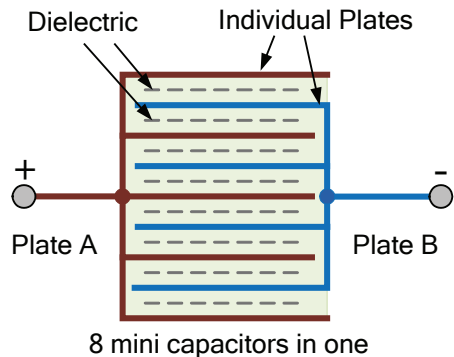
3. MULTIPLATE CAPACITOR

One simple method used to increase the overall capacitance value of a capacitor while keeping its size to a minimum is to “interleave” more conductive plates together within a single capacitor body. Instead of just one set of parallel plates, a capacitor can have many individual plates connected together thereby increasing the surface area, **A** of the plates.

For a standard parallel plate capacitor, it has two plates. Therefore, as the number of capacitor plates is two, we can say that $n = 2$, where “ n ” represents the number of plates.

Thus from the equation above, a single parallel plate capacitor, $n - 1 = 2 - 1$ which equals 1 as $C = (\epsilon_o * \epsilon_r \times 1 \times A)/d$ which is exactly the same as saying: $C = (\epsilon_o * \epsilon_r * A)/d$ above.

FIGURE 3. A MULTIPLATE CAPACITOR



Suppose for a moment we have a capacitor made up of 9 interleaved plates as shown in Figure 3. If there are five plates connected to one terminal (A) and four plates connected to terminal (B), then $n = 9$.

Here, both sides of the four plates connected to terminal B are in contact with the dielectric, whereas only one side of each of the outer top and bottom plates connected to terminal A is in actual contact with the dielectric.

Therefore, the useful surface area of each set of conductive plates will be eight (8) with the capacitance of this multiplate capacitor given as:

The capacitance of the multiplate capacitor example.

$$C = \frac{\epsilon_r \epsilon_o (n-1)A}{d} = \frac{\epsilon_r \epsilon_o (9-1)A}{d} = \frac{\epsilon_r \epsilon_o 8A}{d} \text{ Farads}$$

4. UNITS OF CAPACITANCE

The SI unit of capacitance is the Farad (abbreviated to **F**), named after the British physicist Michael Faraday, is defined as being: “a capacitor has the capacitance of One Farad when a charge of One Coulomb is stored on the plates by a voltage of One volt”.

Capacitance (C), is therefore a measurement of “coulombs per volt” and as such is always positive in value as there are no negative units. However, the Farad “F” unit on its own is a very large unit of measurement for us to use. So prefixes in the form of sub-multiples of the Farad unit are generally used, such as micro-farads, nano-farads and pico-farads. For example.

Prefixes used for Farads (F)

- **Microfarad** (μF) $1\mu\text{F} = 1/1,000,000 = 0.000001 = 10^{-6} \text{ F}$
- **Nanofarad** (nF) $1\text{nF} = 1/1,000,000,000 = 0.000000001 = 10^{-9} \text{ F}$
- **Picofarad** (pF) $1\text{pF} = 1/1,000,000,000,000 = 0.000000000001 = 10^{-12} \text{ F}$

A typical capacitor used in electrical and electronic circuits will have a capacitance in the range of a few picofarads (pF) to hundreds of microfarads (μF). Note however, that today with the increasing demand for energy storage devices, supercapacitors and ultracapacitors are readily available with capacitance values in the ten’s of Farads.

5. STANDARD CAPACITANCE VALUES

Discrete capacitors are commonly available in standard capacitance values depending on their physical size and dielectric material. The capacitance value of a capacitors depends greatly on tolerance, which is based on the same E-Series system as used for the resistors.

For example, the E12 Series (10% tolerance) has nominal values in the multiples of:

1.0 1.2 1.5 1.8 2.2 2.7 3.3 3.9 4.7 5.6 6.8 8.2

So if we take the first value (1.0) and multiply be 10 each time, this would produce a series of standard capacitance values of:

1.0pF, 10pF, 100pF, 1nF, 10nF, 100nF, 1uF, 10uF, 100uF, 1000uF, 10,000uF, etc.

and the same idea of times 10 multiplication for each nominal value in the E12 series is:

1.0, 1.2, 1.5,.. 10, 12, 15,.. 100, 120, 150,.. 1K, 1.2K, 1.5K,.. 10K, 12K, 15K,.. 100K, 120K, 150K,.. and so on.

6. ENERGY STORED IN A CAPACITOR

As a capacitor charges up from an external power supply connected to it, an electrostatic field is created as a result of opposite charges being stored on its two conductive plates. The amount of potential energy, given in Joules, which can be stored in this electrostatic field is equal to the capacitance value of the capacitor and to the square of the voltage supply doing the work to maintain the opposite charges on the plates of the capacitor.

The energy stored by a parallel plate capacitor is given by the formula:

$$W = \frac{1}{2}CV^2 = \frac{CV^2}{2} \text{ Joules (j)}$$

Where: W is the stored energy in joules, C is the capacitance of the capacitor in farads, V is the applied voltage.

Thus the larger the capacitance of the capacitor, or the greater the applied voltage, the more energy a capacitor can store. *Ultracapacitors* and *Supercapacitors* are commonly used as energy storage devices due to their high capacitance values.

Standard capacitors such as electrolytic's are capable of storing electrical energy and charge for long periods of time after their voltage source has been removed, so be careful when touching or handling large value capacitors.

The various names given to describe different types of capacitors generally relate to the

7. CAPACITOR TYPES

type of dielectric material used within its construction. This is because the performance and application of a capacitor is usually dependent upon the dielectric material. There are also both fixed value and variable types of capacitors available which allows us to vary their capacitance value for use in filters or frequency tuning type circuits.

The different types of capacitor available range from small disc and tubular ceramics, to silvered mica, metallised film, or aluminium foil, to those with plastic dielectrics such as polyethylene, Mylar, polypropylene, polycarbonate, and polyester. Larger electrolytic capacitors take the form of Aluminium or Tantalum Electrolytic Capacitors which can be either polarised or non-polarised types.

Figure 4 shows the various schematic symbols used for the different types of capacitor.

FIGURE 4. SCHEMATIC SYMBOLS FOR DIFFERENT CAPACITOR TYPES

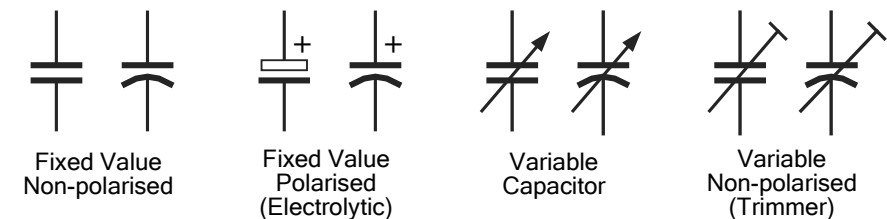


FIGURE 5. DIELECTRIC CAPACITOR


Dielectric Capacitors shown in Figure 5. are usually of the variable type. They change their capacitance value by the variation in the overlapping area of their conductive plates, or by varying the spacing between them.

Air dielectrics are used for larger capacitance values. Trimmers and smaller variable types use very thin mica or plastic sheets as the dielectric between the plates.

The position of the moving plates with respect to the fixed plates determines the overall capacitance value. The capacitance is generally at maximum when the two sets of plates are fully meshed together. High voltage type tuning capacitors have relatively large spacing's or air-gaps between their plates with breakdown voltages reaching many thousands of volts.

FIGURE 6. FILM CAPACITOR


Film Capacitors as shown in Figure 6, are the most commonly available type of capacitor. They consist of a relatively large family of capacitors with the only difference being in their dielectric properties.

Film capacitors are electrostatic capacitors and as such are non-polarised. Film capacitors use polyester (Mylar), polystyrene, polypropylene, or polycarbonate as their dielectric. They are more commonly known as "Plastic Capacitors".

The main advantage of plastic film capacitors compared to impregnated-paper types, is that they can operate at higher temperatures, have smaller tolerance values and high reliability.

Non-polarised film type capacitors are available in capacitance values ranges from as small as 5pF to as large as 100uF depending upon the actual type of capacitor and its voltage rating.

Film capacitors also come in an assortment of shapes and case styles which include oval, round or rectangular. The capacitor is encased in a moulded plastic shell which is then filled and hermetically sealed with epoxy resin. They are available with either Axial and Radial connecting leads.

FIGURE 7. CERAMIC CAPACITOR


Ceramic capacitors or disc capacitors such as that shown in Figure 7 are also electrostatic capacitors made by coating two sides of a small porcelain or ceramic disc with silver oxide and then stacked together to construct the capacitors body.

For very low capacitance values a single ceramic disc of about 3-6mm is used. Ceramic capacitors have a high dielectric constant (High-K) and are available so that relatively high capacitance's can be obtained in a small physical size.

Ceramic capacitors are non-polarised high voltage capacitors generally used as de-coupling or by-pass capacitors with values ranging from a few picofarads (pF) to one or two microfarads, (μ F).

Ceramic types of capacitors generally have a 3-digit code printed onto their body to identify their capacitance value in pico-farads. Generally, the first two digits indicate the capacitors value and the third digit indicates the number of zero's to be added.

For example, a ceramic disc capacitor with the markings 470 would indicate 47 and no zero's in pico-farads which is equivalent to 47pF. Letter codes are also used to indicate their tolerance value such as: J = 5%, K = 10% or M = 20% etc.

Capacitors are available as many different types for a whole range of applications and uses

Electrolytic Capacitors of the type shown in Figure 8 are generally used when very large capacitance values are required. Here instead of using a very thin metallic film layer for one of the electrodes, a semi-liquid electrolyte solution in the form of a jelly or paste is used which serves as the second electrode (usually the cathode).

FIGURE 8. ELECTROLYTIC CAPACITOR



The dielectric is a very thin layer of oxide which is grown electro-chemically in production with the thickness of the film being less than ten microns. This insulating layer is so thin that it is possible to make capacitors with a large value of capacitance for a small physical size as the distance between the plates, d is very small.

The majority of electrolytic types of capacitors are Polarised, that is one plate(s) of the capacitor are designed to store a positive (+ve) charge, while the other plate(s) are designed to store an equal and

opposite negative (-ve) charge.

The DC voltage applied to the capacitor terminals must be of the correct polarity, i.e. positive to the positive terminal and negative to the negative terminal. An incorrect polarisation will break down the insulating oxide layer and permanent damage of the capacitor may result.

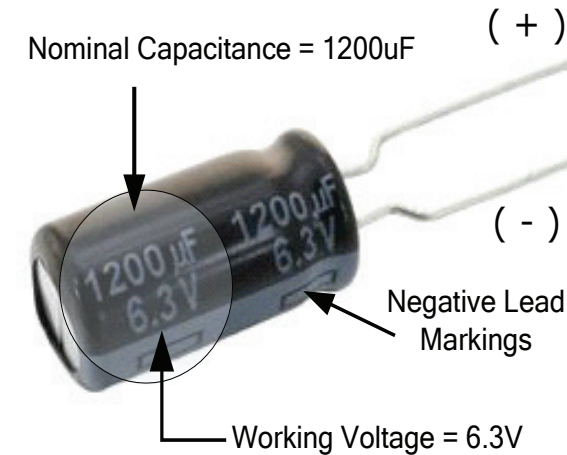
All polarised electrolytic capacitors have their polarity clearly marked with a negative sign used to indicate the negative terminal and this polarity marking must always be followed.

8. CAPACITOR CHARACTERISTICS

Each type or family of capacitors uses its own unique form of identification. Depending upon the manufacturer, some can be easy to understand, such as those shown in Figure 9, while others can use misleading letters, colours or symbols of a non-standard form.

There are a bewildering array of capacitor characteristics and specifications associated with the humble capacitor and reading the information printed onto its body or case can sometimes be difficult to understand especially when colours or numeric codes are used.

FIGURE 9. CAPACITOR MARKINGS



The best way to figure out which capacitor characteristics the label means is to first figure out what type of family the capacitor belongs to whether it is ceramic, film, plastic or electrolytic and from that it may be easier to identify the particular capacitor characteristics.

Even though two capacitors may have exactly the same capacitance value, they may have different voltage ratings. If a smaller rated voltage capacitor is substituted in place of a higher rated

voltage capacitor, the increased voltage may damage the smaller capacitor.

The capacitor, as with any other electronic component, comes defined by a series of characteristics. These Capacitor Characteristics can always be found in the data sheets that the capacitor manufacturer provides to us so here are just a few of the more important ones.

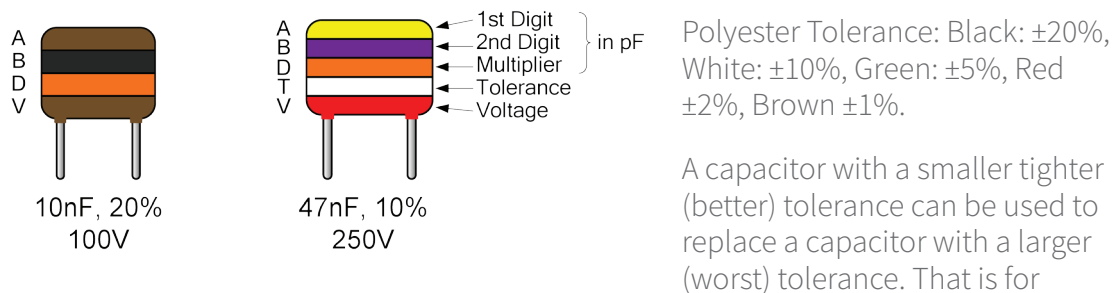
- 1. Nominal Capacitance (C)** - The nominal value of the capacitor given in Farads, (F) is the most important of all the capacitor characteristics. This value can be given in pico-Farads (pF), nano-Farads (nF) or micro-Farads (μ F). It is usually marked onto the body of the capacitor as numbers, letters or coloured bands.
- 2. Working Voltage (WV)** - The working voltage defines the maximum continuous DC voltage that can be applied to the plates of the capacitor without failure of the dielectric. Generally, the working voltage is printed onto the side of a capacitors body with common working DC voltages are 10V, 16V, 25V, 35V, 50V, 63V, etc.

3. **Tolerance ($\pm\%$)** - Capacitors have a tolerance rating expressed as a plus-or-minus value either in picofarad's ($\pm\text{pF}$), for low value capacitors generally less than 100pF or as a percentage ($\pm\%$) for higher value capacitors generally higher than 100pF that the actual capacitance is allowed to vary from its nominal value. For electrolytic capacitors this can range anywhere from -20% to +80% and still remain within tolerance.
4. **Polarization** – This generally refers to electrolytic and tantalum type capacitors with regards to their electrical connection. The majority of electrolytic capacitors are polarized types so must be connected to the correct polarity. Polarized capacitors will have their negative, -ve terminal clearly marked with either a black stripe, band, arrows or chevrons down one side of their body to prevent any incorrect connection to the DC supply.

9. CAPACITOR COLOUR CODING

As well as marking the value and tolerance rating of capacitors with letters and numbers, capacitors can also use a colour coding scheme to identify their values and tolerances. This type of identification consists of coloured bands (in spectral order) as shown.

FIGURE 10. CAPACITOR COLOUR CODING



example, a $\pm 5\%$ (green) tolerance coloured band can be used to replace a $\pm 10\%$ (white) or a $\pm 20\%$ (black) band capacitor.

TABLE 1. CAPACITOR COLOUR BANDS

Colour	Band (A and B)	Multiplier (D)	Tolerance (T)		Voltage (V)		
			(T) $>10\text{pF}$	(T) $<10\text{pF}$	Polyester	Mica	Tantalum
Black	0	x1	$\pm 20\%$	$\pm 2.0\text{pF}$	-	100V	10V
Brown	1	x10	$\pm 1\%$	$\pm 0.1\text{pF}$	100	200V	-
Red	2	x100	$\pm 2\%$	$\pm 0.25\text{pF}$	250	300V	4V
Orange	3	x1000	$\pm 3\%$	-	-	400V	40V
Yellow	4	x10000	$\pm 4\%$	-	400	500V	6.3V
Green	5	x100000	$\pm 5\%$	$\pm 0.5\text{pF}$	-	600V	16V
Blue	6	x1000000	-	-	630	700V	20V
Violet	7	-	-	-	-	800V	50V
Grey	8	x0.01	+80%, -20%	-	-	900V	25V
White	9	x0.1	$\pm 10\%$	$\pm 1.0\text{pF}$	-	1kV	3V

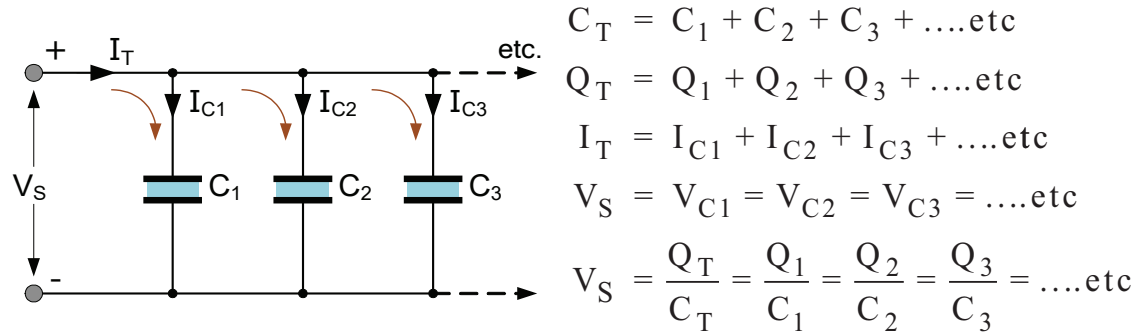
Note that the capacitor colour banding system is used on non-polarised polyester and mica capacitors. While the system of colour coded bands is less common these days, there are still many “old” type capacitors around.

10. CAPACITORS IN PARALLEL

Capacitors are said to be connected together “in parallel” when both of their terminals are respectively connected to each terminal of the other capacitor or capacitors as shown in Figure 11. For capacitors connected together in parallel the supply voltage, (V_s) will always be THE SAME for all parallel connected capacitors.

Thus, capacitors in parallel have a common voltage across them so all the capacitors must charge up to the same voltage level.

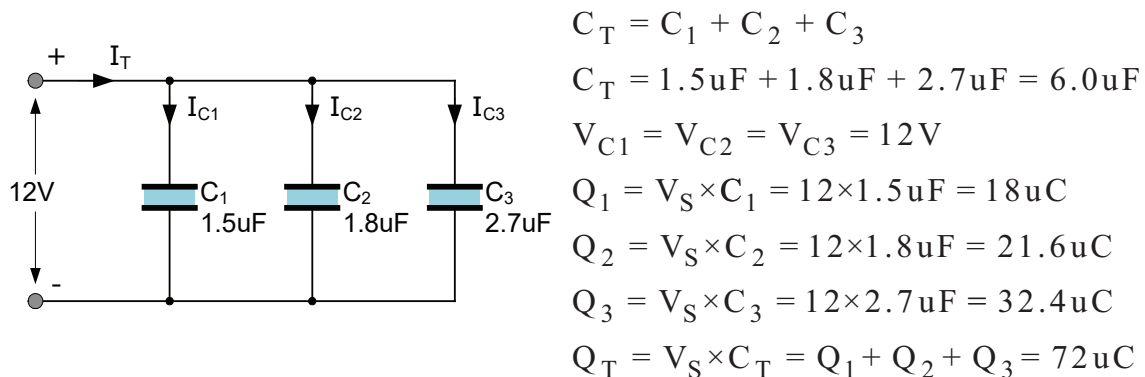
FIGURE 11. CAPACITORS CONNECTED IN PARALLEL



Connecting capacitors together in parallel effectively increases the total area of the conductive plates across the supply voltage making the total capacitance equal to the sum of the individual capacitances. Therefore the equivalent capacitance (C_T) of any two or more capacitors connected together in parallel will always be GREATER than the value of the largest capacitor in the parallel group as we are adding together values.

Parallel connected capacitors can therefore be replaced with one single equivalent capacitor of the same capacitance value. Also, the total charge Q_T stored on all the plates of the equivalent capacitor will equal the sum of the individual stored charges on each of the individual capacitors.

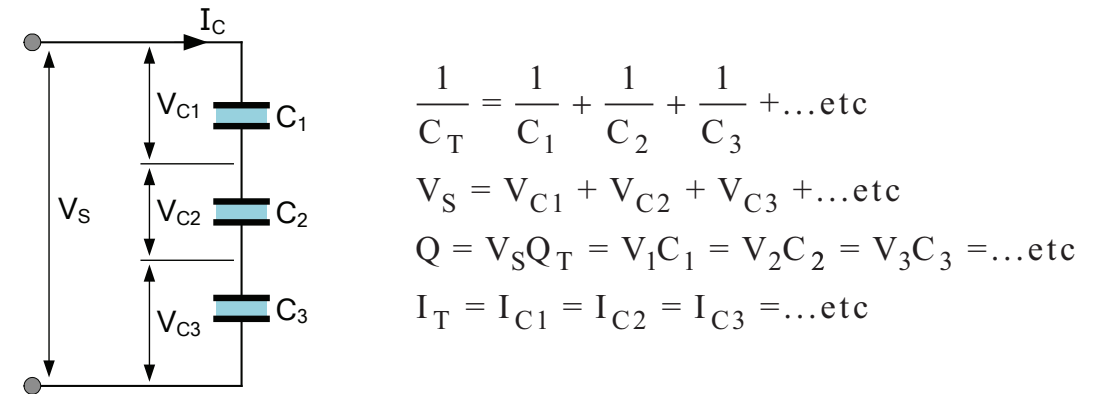
FIGURE 12. PARALLEL CAPACITOR EXAMPLE



10. CAPACITORS IN SERIES

Capacitors are said to be connected together “in series” when they are effectively “daisy chained” together in a single line as shown in Figure 13. Capacitors connected in series effectively increases the thickness of the dielectric, which decreases the total capacitance.

FIGURE 13. CAPACITORS CONNECTED IN SERIES



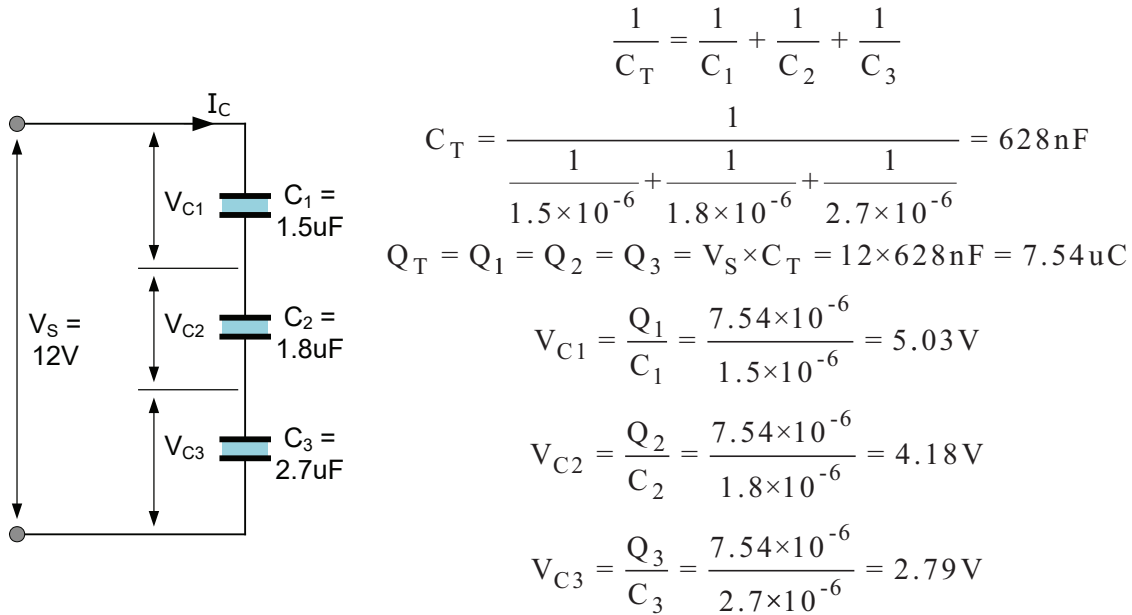
The AC current (I_C) flowing through the capacitors is THE SAME for all capacitors as it only has one path to follow. Thus, capacitors in series all have a common current so each capacitor stores the same amount of charge on its plates regardless of its capacitance. While series capacitors may all have the same alternating current flowing through them as: $I_T = I_{C1} + I_{C2} + I_{C3} + \dots \text{etc}$ there will be a different voltage drop across each of them.

To find the total capacitance value of series capacitors, the reciprocal ($1/C$) of the individual capacitors are added together instead of the capacitance's themselves. Then the total series value equals the reciprocal of the sum of the reciprocals of the individual capacitances. That is:

$$\frac{1}{C_T} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \frac{1}{C_4} + \dots \text{etc}$$

One important point to remember about capacitors that are connected together in a series configuration. The equivalent or total capacitance (C_T) value of any number of capacitors connected together in series will always be LESS than the value of the smallest capacitor in the series chain.

FIGURE 14. SERIES CAPACITOR EXAMPLE

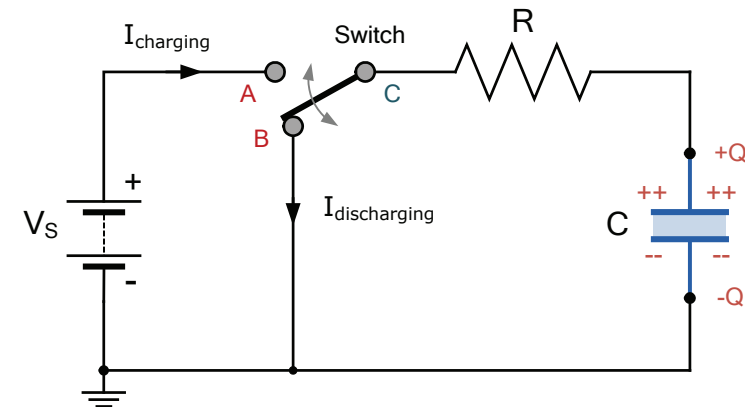


11. RC CHARGING CIRCUIT

A single resistor and a single capacitor can be connected together in series to form an RC (resistor-capacitor) circuit. Resistor-capacitor RC circuits are widely used in a variety of applications such as electronic filters and timing circuits. Delays and timing periods of a few micro-seconds to many hours is possible with the correct selection of components.

The simplest electronic timing circuits are those based on the charging and discharging of a capacitor through a resistor. The charging and discharging of a capacitor can never instant because it takes a certain amount of time to charge or discharge the plates. The time taken for a capacitor to charge or discharge to within a certain percentage of its maximum or minimum value is known as its: **Time Constant (τ)**.

FIGURE 15. AN RC CHARGING/DISCHARGING CIRCUIT



Consider the simple RC circuit shown in Figure 15. Initially, the switch is grounded in position B and not connected to the battery supply, V_S . Thus no current flows through the resistor and the capacitor is uncharged. Then the initial conditions of this RC circuit are given as: $i = 0$ and $q = 0$.

When the switch is moved to position A, time begins at $t = 0$ and current, being defined as a flow of charge in units of amperes, flows into the capacitor via the current limiting resistor, R .

Since the capacitor is fully discharged, the initial voltage across it will be zero, ($V_c = 0$) so the capacitor appears as a short circuit to the supply voltage, V_S .

Capacitors charge and discharge exponentially

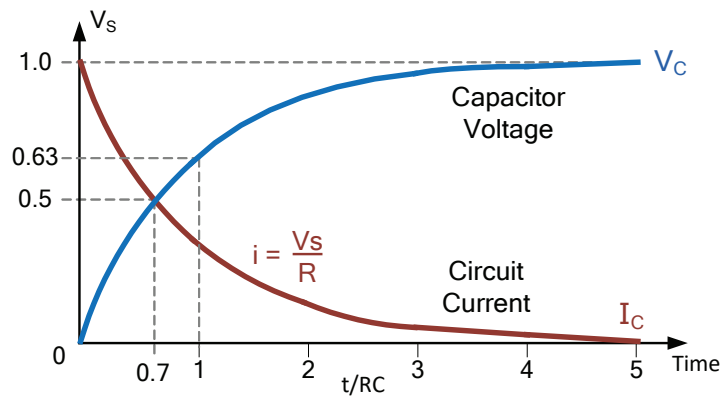
The result is that a current equal to V_S/R flows through the circuit at time $t = 0$ restricted

only by the resistive value R of the resistor. As a result of this instantaneous flow of current, the capacitor begins to charge up storing the charge onto its plates as it does.

As the capacitor charges-up, the voltage across its plates starts to slowly rise and is equal to the charge on it divided by the capacitance, $V = Q/C$. However, since the capacitor and resistor are effectively connected in series, as the voltage across the capacitor rises the voltage across the resistor decreases by an equal amount.

This causes the charging current flowing around the circuit to also reduce as $i = (V_s - V_c)/R$. The charging current continues dropping as the capacitor charges up towards a final value which will be equal to the supply voltage, V_s as shown in Figure 16.

FIGURE 16. RC CHARGING VOLTAGE AND CURRENT CURVES



After a period of time, the voltage across the capacitor, V_c eventually reaches the same voltage level as the supply, V_s and as we can see from the graph this does not happen linearly but follows the exponential mathematical function of: $A \cdot e^{-t/RC}$, where t is the time and RC is the circuits time constant.

The current flowing around the circuit and therefore through the resistor, depends solely on the rate of change of the charge on the capacitor. If at time $t = 0$ the capacitor is completely discharged, the charging voltage derived across the capacitor at any moment in time during charging is defined as:

$$V_{C(t)} = V_s(1 - e^{-t/RC})$$

Then we can see that as the capacitor charges up, the voltage across it rises exponentially, while the circuit current falls exponentially as shown in Figure 16.

As the process of charging a capacitor depends on time, for any given value of “ C ”, the rate at which the capacitor charges will depend on the value of R . Thus the product of R and C produce what is called the Time Constant of the circuit. The *time constant* value of an RC circuit is given the Greek letter, **Tau**, (τ).

Thus one time constant ($\tau = 1RC$) is the exact time it takes for the capacitor to reach 63% ($0.63V_s$) of its fully charged value during charging, and is said to be “fully-charged” after 5 time constants ($5RC$) have passed. This allows us to calculate the capacitor voltage at any given time “ t ” whilst the capacitor is charging.

12. RC DISCHARGING CIRCUIT

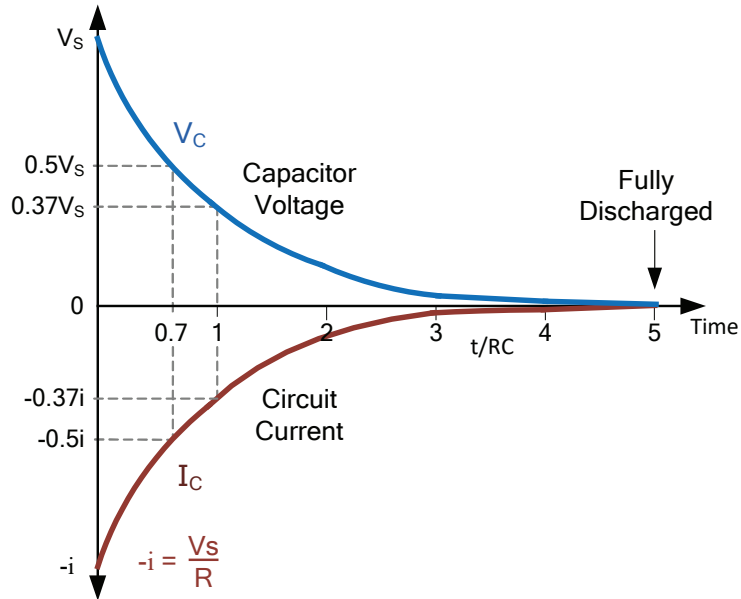
We now assume that the capacitor is fully charged and has been connected to the supply voltage, V_s for a time period greater than 5 time constants, or $5RC$. If the switch above is now moved to position B in Figure 15, the supply voltage is now removed and replaced by a short circuit directly to ground. The capacitor now begins to discharge itself back through the resistor, R in the reverse direction creating an RC discharging circuit.

When the switch is moved to position B, at time $t = 0$, current begins to flow as the fully charged capacitor discharges itself through the now parallel connected resistor, R . Initially the resistor sees all the voltage across the fully charged capacitor, thus $V_R = V_c$ and maximum circuit current flows because $i = V_c/R$.

As the capacitor discharges itself through the resistor, the amount of charge (Q) stored on the capacitors plates reduces so the Q/C voltage across the capacitor also reduces. Thus the capacitor voltage which drives the discharging current becomes less and so the circuit current decreases at a declining rate determined by the RC time constant of the circuit.

Then the flow of current around the circuit initially starts at maximum and gradually reduces to zero when the capacitor is “fully-discharged” as shown in Figure 17.

FIGURE 17. RC DISCHARGING VOLTAGE AND CURRENT CURVES



If the electric charge on the capacitor at any time during its discharge is given by the equation $Q = V \cdot C$, the potential difference across the capacitor at any instant in time as it discharges will be given as:

$$V_{C(t)} = V_s \cdot e^{-t/RC}$$

Thus for an RC discharging circuit, one time constant ($\tau = 1RC$) is the time that it takes for the capacitor voltage to drop to 36.6% of its fully charged value as it discharges. In other words, $1RC$ is the time for a discharging capacitor to lose 63% of its original charge. As the discharging curve for a RC discharging circuit is exponential, for all practical purposes, after five time constants ($5RC$) a capacitor is considered to be fully discharged.

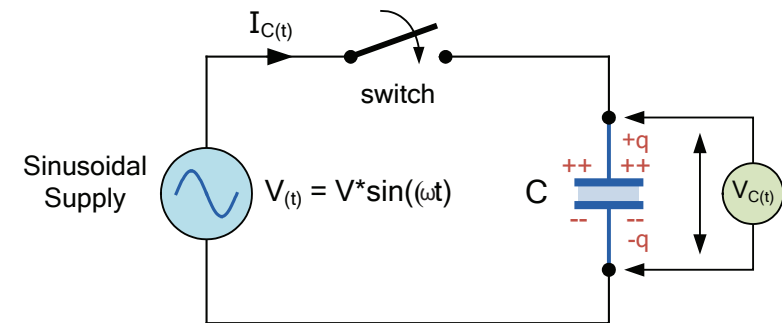
13. CAPACITORS IN AC CIRCUITS

We have seen that capacitors store energy on their conductive plates in the form of an electrical charge and when a capacitor is connected across a DC voltage supply it will charge-up to the value of the applied voltage at a rate determined by its RC time constant.

The relationship between this charging current and the rate at which the capacitors supply voltage changes can be defined mathematically as: $i = C(dv/dt)$, where C is the capacitance value of the capacitor in farads and dv/dt is the rate of change of the supply voltage with respect to time.

However, when connected to an alternating sinusoidal supply voltage, the capacitor will alternately charge and discharge at a rate determined by the frequency of the supply. Then capacitors in AC circuits are constantly charging and discharging respectively. Consider the circuit of Figure 18.

FIGURE 18. CAPACITOR WITH A SINUSOIDAL SUPPLY



When a sinusoidal AC voltage given by the mathematical expression of: $V(t) = V \cdot \sin(\omega t)$ is applied to the plates of an AC capacitor. The capacitor is charged firstly in one direction by the positive half of the sinusoidal waveform, and then in the opposite direction by the negative half of the sinusoidal waveform. Thus the capacitor voltage changes polarity at the same rate as the AC supply voltage.

This instantaneous change in voltage across the capacitor is opposed by the fact that it takes a certain amount of time to deposit (or release) charge onto the plates and for an AC supply is given by: $Q_{(t)} = CV_{(t)} = CV \sin(\omega t)$.

As charge is the flow of electrons, the opposition to the flow of electrons around a circuit is called resistance. In a steady state DC circuit, a capacitor will have an infinite resistance because current can only flow during the capacitors charging and discharging process. However, in an AC circuit in which the plates of the capacitor are constantly charging and discharging, the opposition to current flow is called **Reactance**.

Capacitive Reactance is the opposition to current flow in an AC circuit

Reactance is measured in Ohm's and is given the symbol "**X**". As the component in question is a capacitor, the reactance of a capacitor is called **Capacitive Reactance**, (X_C) which is measured in Ohms. Thus capacitive reactance is the opposition to current flow given by a capacitor in and only in an AC circuit.

CAPACITIVE REACTANCE FORMULA

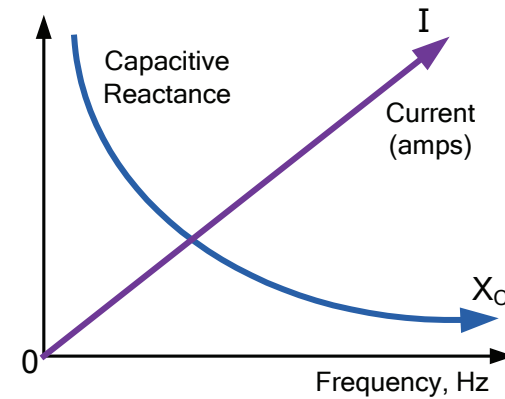
$$X_C = \frac{1}{2\pi fC}$$

Where: X_C is the Capacitive Reactance (in Ohms), f is the supply frequency (in Hertz) and C is the capacitance value of the capacitor (in Farads).

Since capacitors charge and discharge in proportion to the rate of voltage change across them, the higher the supply frequency, the faster the voltage changes, and the more current will flow. This is because there is less time for charge to be stored on the capacitors plates, therefore reducing the capacitor's equivalent resistance.

Likewise, a lower frequency means the slower the voltage changes, the less current will flow. Thus the equivalent resistance of a capacitor (its reactance) is "inversely proportional" to the frequency of the voltage supply as shown in Figure 19.

FIGURE 19. CAPACITIVE REACTANCE AGAINST FREQUENCY



A capacitors capacitive reactance decreases as the frequency across its plates increases. Therefore, capacitive reactance is inversely proportional to frequency.

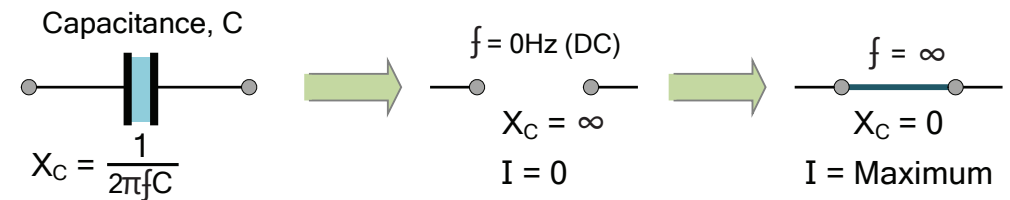
As the frequency increases the AC current flowing through the capacitor increases in value because the rate of voltage change across its plates increases.

Smaller value capacitance means more reactance and more Ohms, therefore less current flow.

Then in an AC circuit, capacitive reactance, X_C depends on the frequency of the supply voltage and the amount of capacitance the capacitor has in farads.

We can show the effect of very low and very high frequencies on the reactance of a pure capacitor as shown in figure 20.

FIGURE 20. EFFECT OF FREQUENCY ON A PURE CAPACITOR



Thus at high frequencies a capacitor acts like a short-circuit, and at low frequencies and DC voltages, a capacitor acts like an open-circuit. As such one common application of a capacitor is as a DC blocking devices preventing direct currents flowing from one electronic circuit or stage to another. For example, an amplifier.

13.1 The Frequency Domain

In an AC circuit containing a pure capacitance, the maximum current flowing into the capacitor is given as:

$$I_{MAX} = \frac{V_{MAX}}{X_C} \quad \text{where: } X_C = \frac{1}{2\pi fC}$$

and therefore, the rms current flowing into an AC capacitance will be defined as:

$$I_{C(t)} = I_{MAX} \sin(\omega t + 90^\circ)$$

Where: $I_C = V/(1/2\pi fC)$ (or $I_C = V/X_C$) is the current magnitude and $\theta = +90^\circ$ which is the phase difference or phase angle between the voltage and current. For a purely capacitive circuit, I_C leads V_C by 90° (ICE), or V_C lags I_C by 90° .

13.2 The Phasor Domain

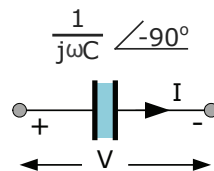
In the phasor domain the voltage across the plates of an AC capacitance will be:

$$V_C = \frac{1}{j\omega C} \times I_C$$

$$\text{where: } \frac{1}{j\omega C} = jX_C = \frac{1}{2\pi fC} = \text{Impeadance, } Z$$

and in Polar Form this would be written as: $X_C \angle -90^\circ$ where:

$$X_C \angle \theta = \frac{V_C \angle 0^\circ}{I_C \angle +90^\circ}$$



Where:

$$X_C \angle \theta = \frac{1}{j\omega C} = 0 - jX_C = \frac{1}{\omega C} \angle -90^\circ = Z \angle -90^\circ$$

End of this Capacitors eBook

Last revision: March 2022

Copyright © 2022 Aspencore

<https://www.electronics-tutorials.ws>

Free for non-commercial educational use and not for resale

With the completion of this Capacitors and Capacitance eBook you should have gained a basic understanding and knowledge of capacitors and how to use them in your circuits. The information provided here should give you a firm foundation for continuing your study of electronics and electrical engineering. In ebook 6 we will learn about inductors.

For more information about any of the topics covered here please visit our website at:

www.electronics-tutorials.ws